Lab 3

BCACC393

Programing with Python Lab

Part A — Lists

1. **Student Marks List**
   Write a Python program to create a list containing the marks of 10 students. Display the maximum mark, minimum mark, total marks, and average marks.
2. **Shopping Basket**
   A customer has a shopping basket containing ["Rice", "Milk", "Bread", "Eggs", "Butter"]. Write a program to add a new item, remove an existing item, and display the final basket.
3. **Temperature Analysis**
   Store the temperatures recorded during 7 days in a list. Write a program to find the highest and lowest temperature and display the days on which they occurred.
4. **Duplicate Removal**
   A list contains repeated roll numbers of students who attended different sessions. Write a Python program to remove duplicate roll numbers and display the unique roll numbers.
5. **List of Even and Odd Numbers**
   Create a list of 20 integers. Write a program to create two separate lists containing the even and odd numbers.
6. **Nested List of Students**
   Create a nested list containing student names and their marks in three subjects. Display each student's name and total marks.
7. **Second Largest Number**
   Write a Python program to find the second-largest number from a list of integers without using the built-in sort() function.
8. **List Rotation**
   Write a Python program to rotate the elements of a list one position to the right. For example, [10, 20, 30, 40] should become [40, 10, 20, 30].

Part B — Tuples

9. **Student Information Tuple**
   Create a tuple containing a student's name, roll number, department, and marks. Display each piece of information using tuple indexing.
10. **Coordinate System**
    A point in a 2D coordinate system is represented as (x, y). Write a program to accept two coordinate tuples and calculate the distance between the two points.
11. **Employee Records**
    Create a tuple containing employee ID, employee name, department, and salary. Display the information using indexing and an f-string.
12. **Tuple Search**
    Create a tuple containing 15 integers. Write a program to accept a number from the user and check whether the number exists in the tuple.

13. **Tuple Statistics**
    Create a tuple containing the marks of 10 students. Write a program to find the maximum, minimum, total, and average marks.
14. **Nested Tuple**
    Create a nested tuple containing information about five books. Each book should contain its title, author, and price. Display the details of all books.

Part C — Dictionaries

15. **Student Marks Dictionary**
    Create a dictionary containing student names as keys and their marks as values. Write a program to display the student with the highest marks.
16. **Contact Book**
    Create a dictionary containing names and phone numbers of five friends. Write a program to search for a friend's phone number using their name.
17. **Product Inventory**
    A shop maintains product names and their quantities in a dictionary. Write a program to add a new product, update an existing quantity, and remove a product.
18. **Word Frequency**
    Write a Python program that accepts a sentence and creates a dictionary containing each word and the number of times it occurs.
19. **Character Frequency**
    Write a Python program to accept a string and create a dictionary containing each character and its frequency.
20. **Employee Salary Dictionary**
    Create a dictionary containing employee names and salaries. Write a program to calculate the average salary and display employees earning more than the average salary.
21. **Subject-Marks Dictionary**
    Create a nested dictionary containing a student's marks in Mathematics, Python, DBMS, and Computer Networks. Calculate and display the total, average, and grade.
22. **Dictionary-Based Result System**
    Create a dictionary containing student names and their marks. Using `if-else`, classify each student as **Distinction, First Class, Pass, or Fail** based on predefined mark ranges.

Learning Outcomes

After completing these laboratory exercises, students will be able to:

1. **Understand and implement** Python data structures such as lists, tuples, and dictionaries for storing and processing data.